

Arjuna JEE 2027

Maths by Amarnath Anand Sir

Quadratic Equations

DPP: 2

Total Duration - 35 Min

- Q1** Find the value of a for which $(a - 2)x^2 + bx + c = 0$ is a quadratic equation.
 (A) for all $a \in R$
 (B) for all $a \in R - \{2\}$
 (C) for all value of a
 (D) for $a \in \{2\}$
- Q2** The value of x satisfying $\log_3(x^2 + 4x + 12) = 2$ are
 (A) 2, -4
 (B) 1, -3
 (C) -1, 3
 (D) -1, -3
- Q3** The number of roots of the equation $x - \frac{2}{x-1} = 1 - \frac{2}{x-1}$ is
 (A) 1
 (B) 2
 (C) 0
 (D) Infinitely many
- Q4** The value of $\sqrt{6 + \sqrt{6 + \sqrt{6 + \sqrt{6 + \dots \infty}}}}$ is
 (A) 3
 (B) $2\sqrt{3}$
 (C) $\sqrt{42}$
 (D) $3\sqrt{2}$
- Q5** The root of the equation $4^x - 3 \cdot 2^{x+2} + 32 = 0$ are
 (A) 1, 2 (B) 1, 3
 (C) 2, 3 (D) 0, 2
- Q6** Let $f(x) = x^2 - 27x + 196$. If $f(x) = x$, then the value of x is
 (A) 28 (B) 14
 (C) 7 (D) 4
- Q7** If one root of the equation $3x^2 - 10x + 3 = 0$ is $\frac{1}{3}$, the other root is
 (A) $-\frac{1}{3}$
 (B) -3
 (C) 3
 (D) $\frac{1}{3}$
- Q8** If $\log_9(x^2 - 5x + 6) > \log_3(x - 4)$, x belongs to
 (A) $(-\infty, 4)$
 (B) $(4, \infty)$
 (C) $(-\infty, -4) \cup (4, \infty)$
 (D) No real value of x
- Q9** The sum of all real values of x satisfying the equation $(x^2 - 5x + 5)^{x^2 + 4x - 60} = 1$ is
 (A) 5 (B) 3
 (C) -4 (D) 6
- Q10** Find the greatest negative integer satisfying $x^2 - 4x - 77 < 0$ and $x^2 > 4$.
- Q11** If $\frac{1}{2} \leq \log_{0.1} x \leq 2$, then
 (A) maximum value of x is $\frac{1}{\sqrt{10}}$
 (B) x lies between $\frac{1}{100}$ and $\frac{1}{\sqrt{10}}$
 (C) minimum value of x is $\frac{1}{\sqrt{10}}$
 (D) minimum value of x is $\frac{1}{100}$
- Q12** For the equation $|x^2| + |x| - 6 = 0$, the roots are
 (A) Real and equal
 (B) Real with sum 0
 (C) Real with sum 1
 (D) Real with sum 2
- Q13** The roots of $15x^2 - 28 = x$ are
 (A) Real and equal
 (B) Real and unequal



- (C) Complex number
- (D) None of these

Q14 The roots of $x^2 - (1 + \sqrt{2})x + \sqrt{2} = 0$ are

- (A) Real and equal
- (B) Real and unequal
- (C) Imaginary
- (D) None of these

Q15 The value of p for which $x^2 + 5px + 16 = 0$ has no real root, are:

- (A) $p > \frac{8}{5}$
- (B) $p < \frac{-8}{5}$
- (C) $\frac{-8}{5} < p < \frac{8}{5}$
- (D) None of these



Answer Key

Q1 (B)
Q2 (D)
Q3 (C)
Q4 (A)
Q5 (C)
Q6 (B)
Q7 (C)
Q8 (B)

Q9 (B)
Q10 -3
Q11 (A, B, D)
Q12 (B)
Q13 (B)
Q14 (B)
Q15 (C)



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